

Assignment 8: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A08_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, lubridate, zoo, and trend packages
 - Set your ggplot theme

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(zoo)
```

```
##
## Attaching package: 'zoo'
```

```
##
## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
```

```
library(lubridate)
library(trend)
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
mytheme <- theme_classic(base_size = 13) +
  theme(axis.text = element_text(color = "black"),
        plot.background = element_rect(fill = "white", color = NA),
        legend.position = "right")
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named GaringerOzone of 3589 observation and 20 variables.

```
#1
epa.air.2010 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv"),
                        stringsAsFactors = TRUE)
epa.air.2011 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv"),
                        stringsAsFactors = TRUE)
epa.air.2012 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv"),
                        stringsAsFactors = TRUE)
epa.air.2013 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv"),
                        stringsAsFactors = TRUE)
epa.air.2014 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv"),
                        stringsAsFactors = TRUE)
epa.air.2015 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv"),
                        stringsAsFactors = TRUE)
epa.air.2016 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv"),
                        stringsAsFactors = TRUE)
epa.air.2017 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv"),
                        stringsAsFactors = TRUE)
epa.air.2018 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv"),
                        stringsAsFactors = TRUE)
epa.air.2019 <- read.csv(here("Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv"),
                        stringsAsFactors = TRUE)
GaringerOzone <- rbind(epa.air.2010,epa.air.2011,epa.air.2012,epa.air.2013,epa.air.2014,epa.air.2015,
                      epa.air.2016,epa.air.2017,epa.air.2018,epa.air.2019)
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.
5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame Days. Rename the column name in Days to "Date".
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame GaringerOzone.

```
# 3
GaringerOzone$Date <- as.Date(GaringerOzone$Date, format = "%m/%d/%Y")
# 4
GaringerOzone1 <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

# 5
Days <- as.data.frame(seq(from = as.Date("2010-01-01"),
                          to = as.Date("2019-12-31"),
                          by = "day"))
colnames(Days) <- "Date"
# 6
GaringerOzone <- left_join(Days, GaringerOzone1, by = "Date")
```

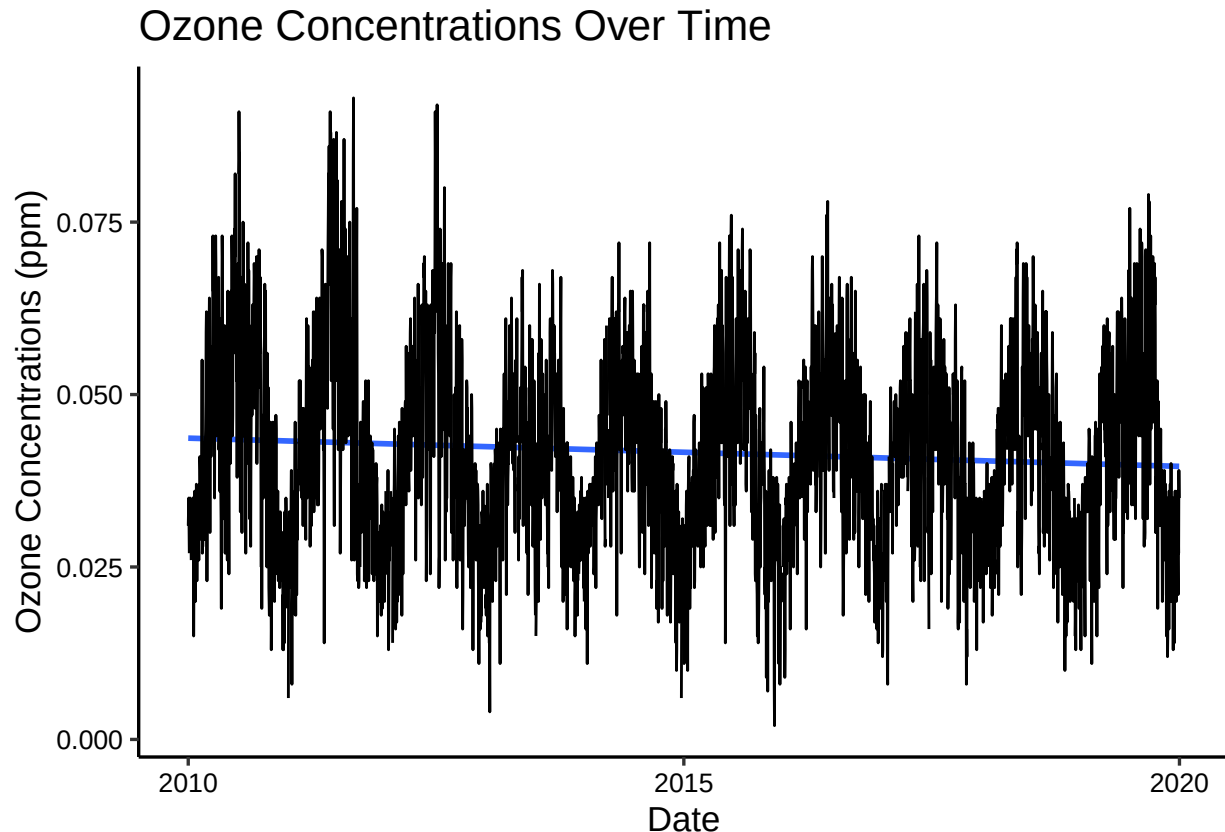
Visualize

7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7
garinger.ozone.plot <- ggplot(GaringerOzone, aes(x = Date,
                                                  y = Daily.Max.8.hour.Ozone.Concentration)) +
  geom_smooth(method = "lm", se = FALSE) +
  geom_line() +
  labs(title = "Ozone Concentrations Over Time",
       x = "Date",
       y = "Ozone Concentrations (ppm)")
print(garinger.ozone.plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```



Answer: My plot suggests that there is a slight decreasing trend in ozone concentration over time.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

```
#8
GaringerOzone <- GaringerOzone %>%
  mutate(
    Daily.Max.8.hour.Ozone.Concentration =
      zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration, x = Date, na.rm = FALSE)
  )
```

Answer: We didn't use a piecewise constant interpolation because that means to carry the last observation forward. It would mean to fill the missing days with existing observations, which is not ideal for atmospheric data. Using spline interpolation would be better for a curved function through points, also not ideal for environmental time series like ozone concentration in the air.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month

to form the groupings. In a separate line of code, create a new Date column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(year = year(Date),
         month = month(Date)) %>%
  group_by(year, month) %>%
  summarize(mean_ozone = mean(Daily.Max.8.hour.Ozone.Concentration,
                             na.rm = TRUE))
```

```
## 'summarise()' has grouped output by 'year'. You can override using the
## '.groups' argument.
```

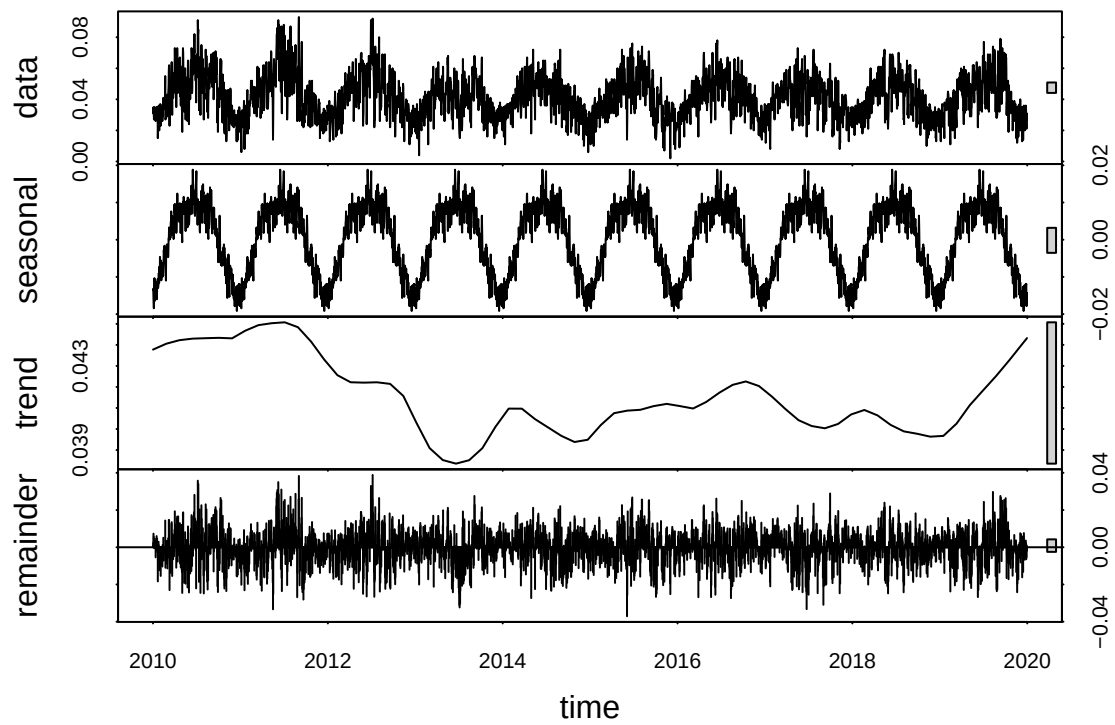
```
GaringerOzone.monthly <- GaringerOzone.monthly %>%
  mutate(MonthYear = make_date(year, month, 1))
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

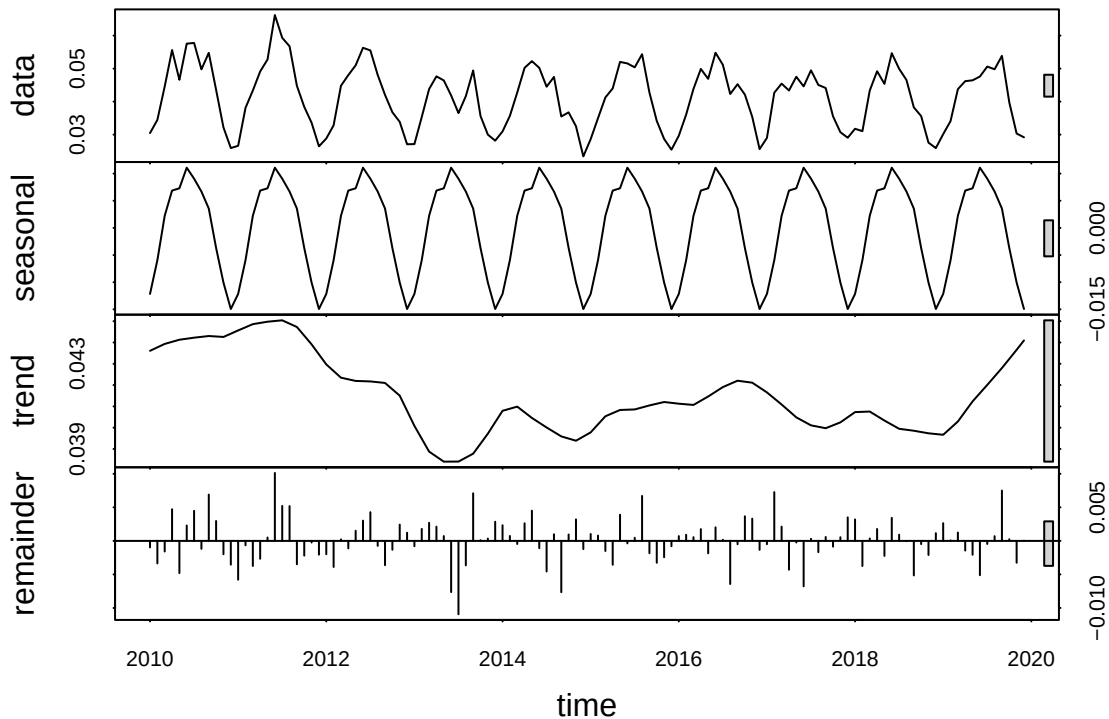
```
#10
GaringerOzone.daily.ts <- ts(
  data = GaringerOzone$Daily.Max.8.hour.Ozone.Concentration,
  start = c(2010, 1),
  end   = c(2019, 365),
  frequency = 365
)
GaringerOzone.monthly.ts <- ts(
  data = GaringerOzone.monthly$mean_ozone,
  start = c(2010, 1),
  end   = c(2019, 12),
  frequency = 12
)
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11
daily.decomposed <- stl(GaringerOzone.daily.ts, s.window = "periodic")
plot(daily.decomposed)
```



```
monthly.decomposed <- stl(GaringerOzone.monthly.ts, s.window = "periodic")
plot(monthly.decomposed)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
ozone.seasonal.trend <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)
summary(ozone.seasonal.trend)

## Score = -77 , Var(Score) = 1499
## denominator = 539.4972
## tau = -0.143, 2-sided pvalue =0.046724
```

Answer: A seasonal Mann-Kendall is most appropriate here because the ozone level data shows a strong seasonal pattern in the sense that the data changes with seasonal cycles. Conducting a seasonal Mann-Kendall helps to reflect that pattern. A regular Mann-Kendall test would assume that the data are independent and not connected on a seasonal level, whereas the seasonal Mann-Kendall would reflect the seasonal cycle consideration.

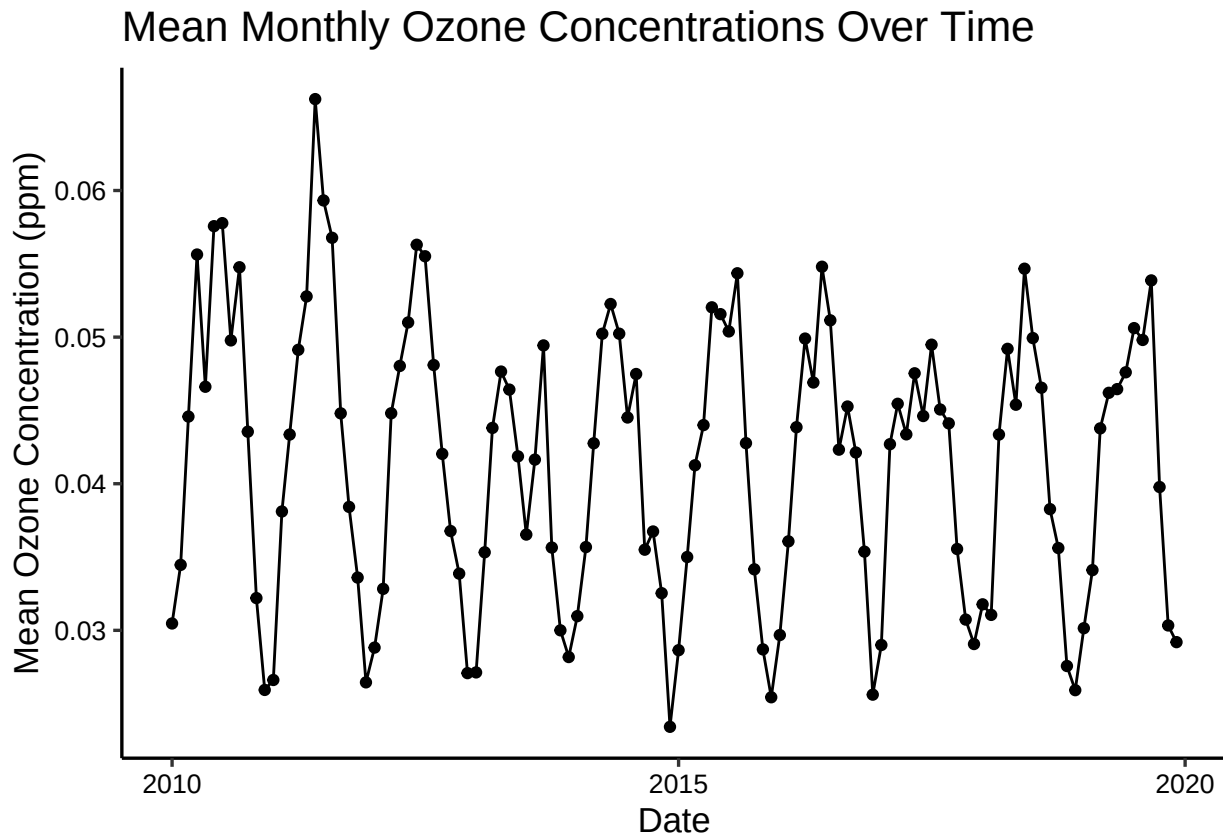
13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
monthly.mean.ozone <- ggplot(GaringerOzone.monthly, aes(x = MonthYear, y = mean_ozone)) +
  geom_line() +
  geom_point() +
```

```

labs(
  title = "Mean Monthly Ozone Concentrations Over Time",
  x = "Date",
  y = "Mean Ozone Concentration (ppm)"
)
print(monthly.mean.ozone)

```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: The mean monthly ozone concentrations at Garinger High School over 2010-2019 show a clear seasonal pattern, indicated by the fluctuating cycle. The overall long-term trend across the time period indicates a gradual decline in ozone levels. The seasonal Mann-Kendall test confirms that there is a downward trend and it is significant ($p < 0.05$)

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.
16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```

#15
GaringerOzone.monthly.deseasoned <- GaringerOzone.monthly.ts - monthly.decomposed$time.series[, 1]

```


#16

```
trend.deseasoned <- Kendall::MannKendall(GaringerOzone.monthly.deseasoned)
summary(trend.deseasoned)
```

```
## Score = -1179 , Var(Score) = 194365.7
## denominator = 7139.5
## tau = -0.165, 2-sided pvalue =0.0075402
```

Answer: The Mann Kendall test result on the non-seasonal Ozone monthly series shows smaller p value and indicates that it is significant ($p < 0.05$). This also shows that there is a decreasing trend in ozone levels (shown from the negative tau) and that despite having the seasonality removed from the series, there still exists the decreasing trend, meaning that the ozone levels have decreased over years despite seasonal changes.